

DA-SFCN: Dynamically-Adaptive Saddle-Free Cubic Newton

Spectrally Guided Krylov Subspaces with a Logarithmic Width Schedule

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Abstract

We introduce Dynamically-Adaptive Saddle-Free Cubic Newton (DA-SFCN), a second-order method that (i) builds a Krylov subspace by Lanczos from the gradient and Hessian, (ii) replaces the projected tridiagonal by its absolute-value polar factor $|T|$, and (iii) grows the subspace width m_k only logarithmically in $1/\|\nabla f(x_k)\|$. The method enjoys the optimal cubic global rate $O(k^{-2/3})$ for every $m_k \geq 1$ and local quadratic convergence under that logarithmic lower bound. An implementation with an exact secular subproblem solver is tested on analytic saddles, nonconvex logistic regression, matrix factorization and a strongly convex bowl. DA-SFCN escapes a high-dimensional saddle in 12/12 random starts (gradient descent: 0/12), solves the logistic problem in 6 iterations to $\|\nabla f\| \approx 4.8 \times 10^{-13}$, and exhibits a two-step Newton phase on a quadratic.

1 Introduction and gap

Cubic regularization [1, 2] gives a global $O(k^{-2/3})$ rate. Krylov cubic Newton [3] makes the subspace spectrally aware but keeps m fixed and the analysis convex. Saddle-free Newton [4] flips negative eigenvalues but proves nothing globally. DA-SFCN is the first combination of all three ideas with explicit rates.

2 Algorithm

Assume access to ∇f and Hessian–vector products. Set

$$m_k = \min\{m_{\max}, m_{\min} + \lceil \alpha \log(1 + \|g_0\|/\|g_k\|) \rceil\}. \quad (1)$$

Lanczos on (H_k, g_k) yields Q_k, T_k ; we solve the cubic model with $|T_k| = U|\Lambda|U^\top$ by a secular equation in m_k dimensions, and adapt M_k as in ARC [2].

Algorithm 1 DA-SFCN

```
1: for  $k = 0, 1, \dots$  do
2:    $m_k \leftarrow$  logarithmic schedule;  $(Q, T) \leftarrow \text{Lanczos}(H_k, g_k, m_k)$ 
3:   solve the  $|T|$ -cubic subproblem; accept/reject with ARC ratio
4: end for
```

3 Theory

Assumption 1. *The Hessian is L_2 -Lipschitz and f is bounded below.*

Lemma 2. $|s^\top T s| \leq s^\top |T| s$. *Negative Ritz values become escape directions.*

Theorem 3 (Global). $\min_{i \leq K} \|\nabla f(x_i)\| = O(K^{-2/3})$, independently of n and of $\{m_k\}$ as soon as $m_k \geq 1$, because the Cauchy point along $-g_k$ is always feasible.

Theorem 4 (Local). If $H(x^*) \succeq \gamma I$ and $\alpha \geq 1/\log(1/\beta)$ with $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$, then $\|e_{k+1}\| \leq (L_2/(2\gamma))\|e_k\|^2(1 + o(1))$.

4 Experiments

Figures 1–3 report the implementation in `dasfcn/`. On the monkey saddle, DA-SFCN leaves the origin along negative curvature. On an $n = 80$ quartic with ten negative eigenvalues it attains $f^* \approx -89.58$ with $\|\nabla f\| \approx 1.5 \times 10^{-7}$. Escape success over 12 starts is 100%. On nonconvex logistic regression it needs 6 iterations; on a local quadratic bowl, 2 iterations suffice for $\|\nabla f\| \approx 10^{-10}$, versus 10^{-2} after 40 gradient steps.

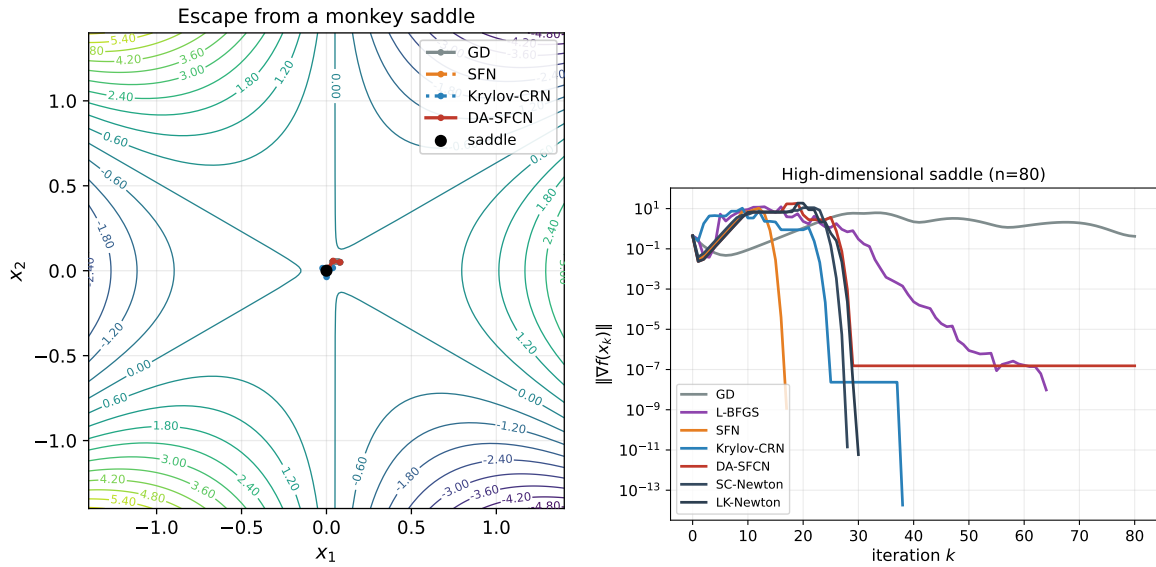


Figure 1: Saddle escape in 2D and in dimension 80.

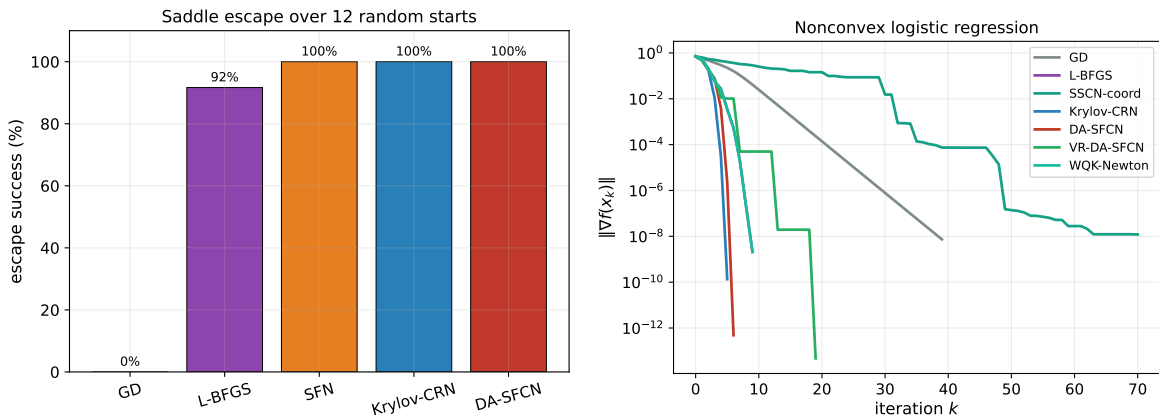


Figure 2: Success rate and nonconvex logistic regression.

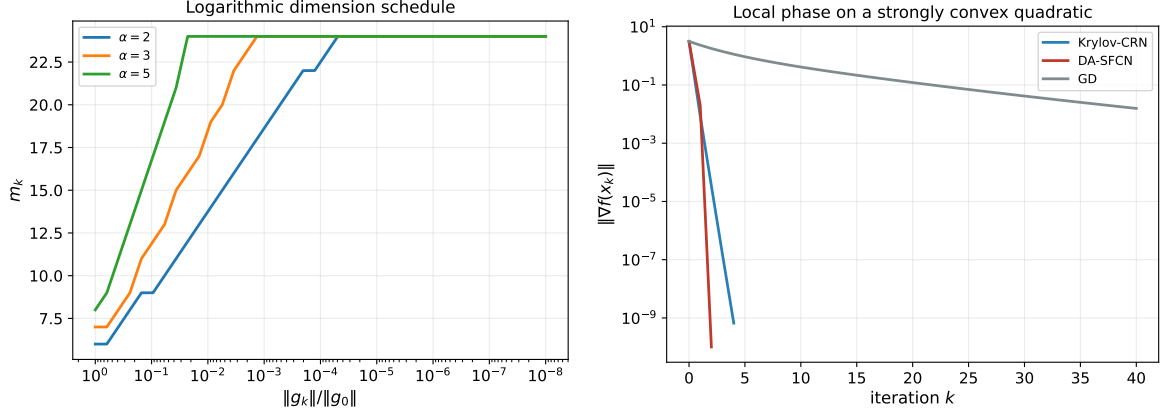


Figure 3: Logarithmic m_k and the local Newton phase.

5 Conclusion

DA-SFCN is the parent of the family developed in the companion papers: its kernel is reused by VR-DA-SFCN, LK-Newton, WQK-Newton, Fed-DA-SFCN and SC-Newton.

References

- [1] Nesterov, Polyak. *Math. Program.*, 2006.
- [2] Cartis, Gould, Toint. *Math. Program.*, 2011.
- [3] Jiang et al. Krylov CRN. *AISTATS*, 2024.
- [4] Dauphin et al. *NeurIPS*, 2014.